

Kristian Dave Ramirez

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WORK EXPERIENCE

Samsung Research Philippines

June 2022 – Present

Python Associate Engineer | March 2025 – Present

Taguig, Metro Manila Philippines

- Promoted due to good technical performance
- Develops and maintains internal AI web applications which performs classification, sentiment analysis, and summarization on thousands of posts daily using Samsung's LLM, Samsung Gauss.
- Currently developing a new system which performs text embedding for clustering and classification using PostgreSQL with pgvector as vector DB and Gemini embeddings as embedding model.
- Maintaining a vLLM server which runs a quantized version of Mistral small 3.1 which is used for text summarization by the new system.

Java Backend Engineer | June 2022 – March 2025

- Implemented and maintained backend server applications using Spring Boot applications on EC2 instances
- Worked on the creation of an ETL Pipeline in Python using Apache Airflow deployed on AWS MWAA
- Passed 2nd level of internal algorithm certification (SWC Pro)

AI Evaluation Engineer | November 2023 – December 2024

- Evaluated and added support for state of the art multimodal models on lmms-eval and VLMEvalKit
- Created automation scripts for benchmark execution using lmms-eval for an internal website.
- Evaluated state of the art image generation models on HPSv2 benchmark and other benchmarks.
- Evaluated speech separation, speech diarization, and speech recognition on benchmarks and test datasets
- Made use of A100 Cloud GPUs for development and execution of benchmarks
 - **Tech Stack:** Python, PyTorch; lmms-eval; VLMEvalKit; huggingface; transformers; datasets

PERSONAL PROJECT

PodQA - a RAG chatbot application

<https://github.com/rafal01010/PodQA>

- A RAG chatbot application using transcripts of Trash Taste, a YouTube podcast, as the knowledge base
- Whisper Turbo is used to perform automatic speech recognition (ASR) to produce the transcripts as SRT files
- LanceDB is used as the vector database to store the embeddings and for ANN retrieval
- GTE ModernBERT and GTE ModernColBERT are used to produce the embeddings. System can be configured to retrieve using dense embeddings or using multivector embeddings from the late interaction model
- Can be configured to use the Gemma 3 4B model to run locally or the Gemini Flash API for faster inference
- Execution and development of the system is done on a base M4 Mac mini with 32 GB of unified memory

EDUCATION

University of the Philippines Diliman

July 2021

Bachelor of Science in Computer Engineering

Quezon City, Philippines

SKILLS & INTERESTS

- **Skills:** Proficiency in Java, Python; Proficiency in Spring Boot and AWS Services; Strong communications skills used in discussing complex engineering topics; Proficiency in Deep Learning and Machine Learning for NLP; Familiarity with RAG, Computer Vision, Convolutional Neural Networks, Diffusion models
- **YouTube Lecture Playlists watched:** Deep Learning for Computer Vision from U. of Michigan, Stanford CS224N, Stanford CS25 series: Emergent Abilities and Scaling in LLM, Intuitions on Language Models, Retrieval Augmented Language Models, Aligning Open Language Models, Stanford NLP Seminar on evaluation of LLMs by Jason Wei, Low Level Technicals of LLMs by Daniel Han, Beyond the Basics of RAG by Ben Clavié
- **Technical Books Read:** AI Engineering by C. Huyen, Build a Large Language Model by S. Raschka
- **Interests:** IZ*ONE, Orb: On the Movements of the Earth; Odd Taxi; REAL; Slam Dunk